

Neural Spectral Band Generation for Audio Coding

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Abstract

Spectral band replication (SBR) enables bit-efficient coding by generating high-frequency bands from the low-frequency ones. However, it only utilizes coarse spectral features upon a subband-wise signal replication, limiting adaptability to diverse acoustic signals. In this paper, we explore the efficacy of a deep neural network (DNN)-based generative approach for coding the high-frequency bands, which we call neural spectral band generation (n-SBG). Specifically, we propose a DNN-based encoder-decoder structure to extract and quantize the side information related to the high-frequency components and generate the components given both the side information and the decoded core-band signals. The whole coding pipeline is optimized with generative adversarial criteria to enable the generation of perceptually plausible sound. From experiments using AAC as the core codec, we show that the proposed method achieves a better perceptual quality than HE-AAC-v1 with much less side information.

Index Terms: audio coding, spectral band replication, generative adversarial training

1. Introduction

The primary objective of audio coding (or audio compression) is to reduce the amount of data needed to represent audio signals while preserving the quality of the decoded output [1, 2]. Audio coding schemes can be classified as either lossless or lossy. Lossless coding preserves the original signal exactly, whereas lossy coding achieves higher compression by permitting some quality loss. In particular, perceptual audio coding [2, 3], a form of lossy compression, aims to maximize the data efficiency while maintaining perceived audio quality as much as possible. These methods exploit psychoacoustics [4]—studies on the relationship between acoustic stimuli and human auditory perception—to determine the optimal bit allocation over frequency bins under a bit-budget restriction. By employing psychoacoustic models (PAMs) to assess the perceptual saliency of audio components, perceptual audio codecs assign fewer bits to less perceptually significant components, thereby achieving a higher compression ratio without noticeably degrading quality.

At low bit-rates, many perceptual audio codecs [5, 6] opt to totally discard some spectral components above a certain frequency, prioritizing the encoding of lower-frequency content. While this strategy efficiently reduces bit-rate requirements, the resulting bandwidth limitation can lead to muffled sound quality [7, Chapter 5]. To address these issues, many perceptual

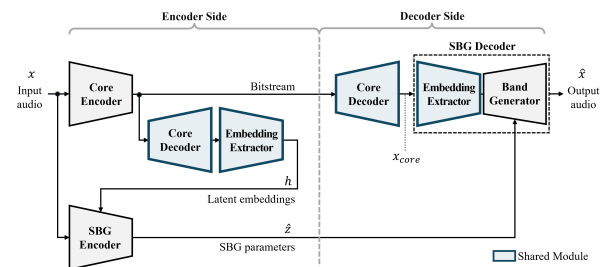


Figure 1: An overview of neural Spectral Band Generation (n-SBG). The SBG encoder extracts quantized parameters from the input audio, while the SBG decoder generates high-frequency subbands based on the transmitted parameters and coded low-frequency bands. The core decoder and embedding extractor modules are shared across the encoder and decoder sides, while the core encoder and core decoder remain fixed and not trained.

codecs such as HE-AAC [8], MP3Pro [9] and Opus [10] incorporate spectral band replication (SBR) [11] (or a similar method), a technique designed to perceptually encode the high-frequency components using the existing core-band signals in an efficient manner. In the encoding stage, the input audio signal is analyzed by a filterbank and the key encoding parameters such as spectral envelope information and noise level estimates are extracted by a SBR encoder to be used for reconstructing the high-frequency components from the low-frequency ones. These parameters are then quantized and transmitted to the decoder. In the decoding stage, the core codec output is separated into subband signals, and a SBR decoder reconstructs high-frequency components by replicating low-frequency subband signals into the high-frequency range. The replicated subband signals are then adjusted based on the transmitted parameters and further enhanced with sinusoids or noise if needed.

The SBR is a well-established parametric approach to audio bandwidth extension (BWE), a task of reconstructing missing high-frequency components from band-limited signals. Specifically in the case of SBR, low-frequency band information along with encoded side information is utilized to generate high-frequency spectral content. In parallel, numerous DNN-based BWE [12–15] approaches have emerged, with the aim of generating missing high-frequency spectra from low-band inputs (i.e., blind BWE). Building upon these approaches, one may expect to replace the SBR with a DNN-based BWE model in the audio coding pipeline. However, this straightforward integration may be unsuitable for general audio signals, as the correlation between the low-band and the high-band characteristics varies depending on the content [16]. Unlike these blind approaches, where the missing high-frequency content must be

This work was supported by Electronics and Telecommunications Research Institute (ETRI) grant funded by the Korean government. [25ZC1100, The research of the basic media-contents technologies]

inferred without prior information, audio coding has access to the full-band signal at the encoder [7, Chapter 5]. This fundamental difference allows for the extraction of extra information that guides high-frequency reconstruction more accurately.

Inspired by SBR, we propose **neural Spectral Band Generation (n-SBG)** as a new framework to integrate DNN-based BWE models into traditional audio coding pipelines for more efficient high-frequency restoration. Figure 1 provides an overview of the proposed n-SBG framework. During encoding, a feature map is extracted and quantized from the original full-band audio, then transmitted as SBG parameters. The decoder utilizes these parameters together with the core codec’s band-limited output to generate the desired high-frequency bands. The key contributions of this research can be summarized as follows:

- We propose a novel framework for parametric coding of high-frequency bands at low bit-rate conditions, replacing the rule-based SBR with a DNN-based encoder-decoder architecture.
- Our work lays the foundation for a new paradigm in high-frequency coding by leveraging information from core codec outputs for more efficient feature extraction and accurate reconstruction.

2. Related Works

2.1. Bandwidth Extension

DNN-based blind BWE approaches typically use generative models such as diffusion models [12, 13] or generative adversarial networks (GAN) [14, 15] to estimate high-frequency content from low-frequency inputs, based on distributions learned from full-band audio data. These methods have been found to be particularly effective for speech signals [15], as high-frequency patterns in speech are more predictable than those in music and general audio, but their performance for general audio has not yet been fully validated.

2.2. Neural Audio Codecs

Neural audio codecs (NACs) compress signals through an encoder-quantizer-decoder pipeline, often incorporating residual vector quantization (RVQ) to discretize latent embeddings from the encoder. Representative examples such as SoundStream [17], EnCodec [18], and DAC [19] employ convolutional autoencoders to process time-domain audio signals. To further improve perceptual quality, these methods often incorporate adversarial training with frequency-domain discriminators. Recently, a blind BWE module was integrated into a NAC to reduce bit-rate [20], primarily focusing on speech signals. To generalize across diverse audio signals, our approach transmits additional parameters for high-frequency generation through a dedicated encoder module.

2.3. Post-Processing with Auxiliary Information

While many works aim to enhance degraded codec outputs solely based on the decoded signal itself [21–23], some post-processing modules incorporate features obtained from the input signal before encoding to improve the performance of the enhancement. For instance, [24] and [25] extract quantized features from the frequency-domain representation of the input signal using convolutional networks, which are then provided for the enhancement. However, these methods primarily focus on speech signals with low sampling rates.

3. Proposed Method

The proposed method consists of two main components—SBG encoder and decoder—as shown in Figure 2. The SBG encoder extracts SBG parameters (lying in an embedding space) from the STFT coefficients of the input audio, while the SBG decoder generates high-frequency subband signals for Pseudo-Quadrature Mirror Filters (PQMF) [26] based on the transmitted SBG parameters and the decoded low-frequency subband signals from the core-codec. Additionally, the output embeddings from the bottleneck of the SBG decoder are provided to the SBG encoder as a conditional input, making the extraction of side information dependent on the coded core bands. Compared to SBR, our method replaces the rule-based parameter extraction and synthesis pipeline with a fully neural-network-based approach, optimized in an end-to-end manner. The details of each processing module are explained in the following subsections.

3.1. SBG encoder

The SBG encoder is designed with an architecture similar to ResNet-18 [27]. The input audio signal $x \in \mathbb{R}^{T'}$ is transformed into a log-power spectrogram $x_{\text{stft}} \in \mathbb{R}^{1 \times F \times T}$, where T' , T , and F indicates the length of the input audio signal, the number of frames, and the number of frequency bins, respectively. Only the frequency bins corresponding to the range generated by the SBG decoder are provided to the feature encoder. The selected spectral coefficients, represented as $\hat{x}_{\text{stft}} \in \mathbb{R}^{1 \times F' \times T}$, are then processed by the feature encoder.

The feature encoder (shown in Figure 2.a) consists of an initial 2D-convolution with a kernel size of (7, 7), stride factor of (2, 1) and an output channel size of $D/8$, followed by max pooling layer with a kernel size of (3, 3) and stride factor of (2, 1), and four *Residual Stages*. Each *Residual Stage* contains 3×3 convolutions and ReLU activations. The last three stages progressively double the input channel size and halve the frequency resolution. The feature encoder then produces an output embedding $z \in \mathbb{R}^{D \times (F'/S_f) \times T}$, where $S_f = 32$. All convolutional layers in the feature encoder are causal. The extracted feature embedding z is linearly projected into an S_f -dimensional space, and then reshaped into $z' \in \mathbb{R}^{F' \times T}$.

Following the quantization scheme of DAC [19], we use RVQ to quantize z' into SBG parameter $\hat{z}$. The RVQ consists of N_q layers of vector quantization (VQ), each with an N -dimensional codebook containing M learnable code vectors, where $N < F'$. The bit-rate of the SBG parameter is calculated as $\frac{f_s}{H} \cdot N_q \cdot \lceil \log_2 M \rceil$ (bps), where f_s denotes the sampling rate and H is the hop length of the STFT.

3.2. SBG decoder

As shown in Figure 2.b, the SBG decoder receives two inputs: (1) the coded core-band signal x_{core} , which is decomposed into critically-sampled subbands ($b_1^{(\text{core})}, \dots, b_{N_{\text{core}}}^{(\text{core})}$) and (2) the SBG parameters $\hat{z}$ from the SBG encoder. Using these, the SBG decoder generates high-frequency bands ($b_{N_{\text{core}}+1}^{(\text{gen})}, \dots, b_{N_{\text{core}}+N_{\text{HF}}}^{(\text{gen})}$), which are subsequently combined with the subbands of the coded core-bands and synthesized into the bandwidth-extended output $\hat{x}$. We use 32-channel PQMF filterbanks for the subband analysis and synthesis.

We adopt SEANet [14, 28], a time-domain speech BWE model, as the backbone for the SBG decoder and extend its architecture to process multi-channel input and output. Specifically, the SBG decoder begins with an initial 1D-convolution

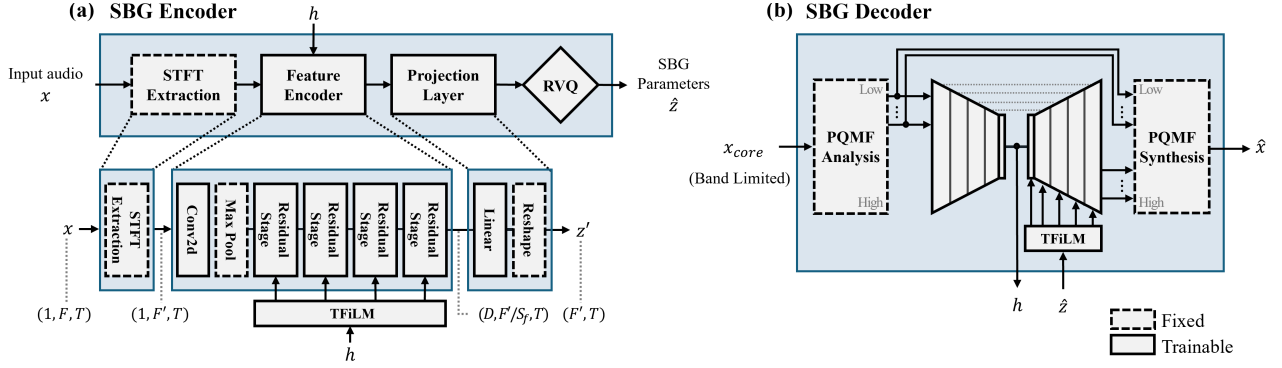


Figure 2: Detailed descriptions of the encoder and the decoder architecture: (a) SBG encoder architecture with details of the feature encoder and the projection layer; (b) SBG decoder architecture.

layer that expand the channel size from N_{core} to C . Next, the feature maps pass through four *Encoder Blocks* (i.e., Embedding Extractor), which downsample along the temporal axis and double the channel size. Following the *Encoder Blocks*, the signal reaches the bottleneck stage, which consists of two 1D-convolution layers. The first layer reduces the channel size from $16C$ into $4C$, and the second layer expands it back to $16C$. Then, four *Decoder Blocks* (i.e., Band Generator) successively invert the process of the *Encoder Block*. An addition-based skip connection links each corresponding encoder and decoder block. Finally, the last 1D-convolution outputs N_{HF} channels corresponding to the high-frequency subbands to be reconstructed. We set the stride factors of the downsampling layers to $(1, 2, 2, 2)$. Given that the PQMF analysis process already downsamples the input signal by a factor of 32, the downsampling layers further reduces the time resolution by an additional factor of $2^3 = 8$, resulting in an overall reduction up to 256. All convolution layers within the SBG decoder are causal, ensuring real-time processing that remains consistent with the causal structure of the SBG encoder.

3.3. Conditioning scheme

The n-SBG leverages two complementary forms of conditioning. First, the SBG decoder extracts latent embeddings $h \in \mathbb{R}^{4C \times (T'/256)}$ from its bottleneck and feeds it to the four *Residual Stages* of the SBG encoder. This makes the extraction of SBG parameters dependent on the coded core band, resulting in more efficient usage of the bit-rate. Second, the SBG parameter $\hat{z} \in \mathbb{R}^{S_f \times (T'/H)}$ is provided to the last bottleneck layer and the four *Decoder Blocks* of the SBG decoder, facilitating more accurate reconstruction of high-frequency bands.

We utilize Temporal Feature-wise Linear Modulation (TFiLM) [29] to modulate the activation of a specific layer at each timestep, conditioning it based on a conditional input. Let $a \in \mathbb{R}^{C_a \times \dots \times T_a}$ be the activation to be modulated, where C_a is the channel size and T_a is the number of timesteps. We extract two parameters $\beta, \gamma \in \mathbb{R}^{C_a \times T_a}$ from a conditional input $b \in \mathbb{R}^{C_b \times T_b}$. If $T_b > T_a$, we use a strided convolution; otherwise, we replicate each timestep of b to align the temporal resolution with a . The resampled tensor $b_{\text{re}} \in \mathbb{R}^{C_b \times T_a}$ is then mapped to the shape (C_a, T_a) through a point-wise linear projection. Finally, each timestep of the activation a is modulated as follows:

$$a_t^{(\text{mod})} = \gamma'_t \cdot a_t + \beta'_t, \quad \text{for } 0 \leq t < T_a, \quad (1)$$

where $a_t, a_t^{(\text{mod})} \in \mathbb{R}^{C_a \times \dots \times 1}$, and $\beta_t, \gamma_t \in \mathbb{R}^{C_a \times 1}$ are reshaped into $\beta'_t \in \mathbb{R}^{C_a \times \dots \times 1}$ and $\gamma'_t \in \mathbb{R}^{C_a}$.

4. Experiments

4.1. Training Objectives

We apply a GAN framework [17–19] to train the n-SBG, treating the entire framework as a generator. The training objective of the generator ($\mathcal{L}_G^{\text{total}}$) is composed of a weighted sum of various loss functions:

$$\mathcal{L}_G^{\text{total}} = \lambda_{\text{mel}} \mathcal{L}_{\text{mel}} + \lambda_{\text{adv}} \mathcal{L}_G^{\text{adv}} + \lambda_{\text{fm}} \mathcal{L}_{\text{fm}} + \lambda_{\text{cb}} \mathcal{L}_{\text{cb}} + \lambda_{\text{cm}} \mathcal{L}_{\text{cm}}, \quad (2)$$

where the weighting coefficients are set as $\lambda_{\text{mel}} = 15$, $\lambda_G^{\text{adv}} = 3$, $\lambda_{\text{fm}} = 6$, $\lambda_{\text{cb}} = 1$, and $\lambda_{\text{cm}} = 0.5$. The multi-scale mel-reconstruction loss, $\mathcal{L}_{\text{mel}}$, is computed across seven different frequency resolutions [19] and is defined as follows:

$$\mathcal{L}_{\text{mel}} = \sum_{i=1}^7 \|\log_{10} M_i(x) - \log_{10} M_i(\hat{x})\|_1, \quad (3)$$

where $M_i(\cdot)$ represents the mel-spectrogram at scale i , computed using a window length of 2^{4+i} , a hop size of 2^{2+i} , and 5×2^i mel bins. For adversarial training, we use the hinge loss ($\mathcal{L}_G^{\text{adv}}$) [30] and the feature matching loss ($\mathcal{L}_{\text{fm}}$). Additionally, we use two auxiliary losses to train the RVQ: the codebook loss ($\mathcal{L}_{\text{cb}}$) and the commitment loss ($\mathcal{L}_{\text{cm}}$) [31]. We employ multi-band STFT-based discriminators [19] and multi-period discriminators [32] to enable high-fidelity reconstruction. These discriminators are trained using a hinge loss ($\mathcal{L}_D^{\text{adv}}$) with a weighting factor of $\lambda_D^{\text{adv}} = 1$.

All training objective functions are calculated between the output signal $\hat{x}$ and the target signal x_{tgt} , where x_{tgt} is synthesized from $(b_1^{(\text{core})}, \dots, b_{N_{\text{core}}}^{(\text{core})}, b_{N_{\text{core}}+1}^{(\text{input})}, \dots, b_{N_{\text{core}}+N_{\text{HF}}}^{(\text{input})})$ via a PQMF synthesis filterbank.

4.2. Dataset and Experimental Setup

For training, we utilize three datasets: FSD-50K [33], MUSDB18-HQ [34], and VCTK [35]. These datasets provide approximately 70 hours of diverse sound events, 30 hours of music, and 30 hours of speech content, respectively. For evaluation, we used 43 candidate test items for USAC standardization [36], consisting of 11 for speech, 18 for music, and 14 for mixed signals, respectively. All datasets have a sampling rate of 48 kHz.

In the experiment, we compare n-SBG with Fraunhofer FDK HE-AAC v1¹, where both share the same core codec. We used HE-AAC v1 at 12 and 16 kbps bitrate settings. The

¹<https://tsrac.ffmpeg.org/wiki/Encode/AAC>

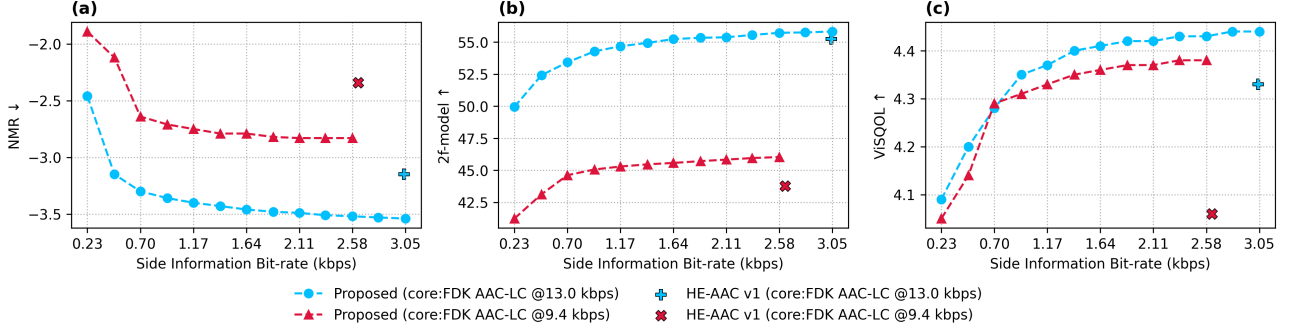


Figure 3: Rate-distortion curves comparing FDK HE-AAC v1 and the proposed model, both generated using the AAC core output at the same side information bitrate. Distortion measure: (a) NMR ↓ (b) 2f-model MMS ↑ (c) ViSQOL MOS ↑

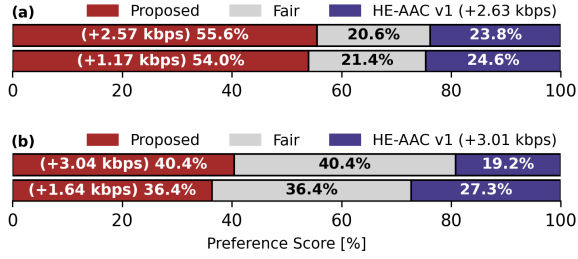


Figure 4: A/B preference test results comparing n-SBG with HE-AAC v1 under different core bit-rate conditions: (a) AAC-LC @9.4 kbps (b) AAC-LC @13.0 kbps. Values in the parentheses indicate the bit-rate of side information.

split of the subbands ($N_{\text{core}}, N_{\text{HF}}$) for the n-SBG follows the configuration of HE-AAC v1, using (5, 10) and (5, 11) at 12 and 16 kbps respectively. The n-SBG encoder outputs a 512-dimensional vector per STFT frame, with both window length and hop length set to 2048, i.e., $(D, H) = (512, 2048)$. RVQ of n-SBG encoder has a maximum bit-rate similar to that of the SBR in HE-AAC v1. Each codebook contains 1024 number of 8-dimensional code vectors, where $N_q = 11, 13$ for 12 and 16 kbps setups, respectively. For inference, the bit-rate of the SBG parameters can be adjusted by bypassing the last few VQ layers. The n-SBG decoder has a channel size of $C = 64$. We trained n-SBG using the Adam optimizer with an initial learning rate of 1.0×10^{-4} , $(\beta_1, \beta_2) = (0.5, 0.9)$, and exponential learning rate decay with $\gamma = 0.999996$ for both the generator and the discriminators.

4.3. Experimental Results

We evaluate the performance of the proposed method using NMR [37], 2f-model MMS [38], and ViSQOL MOS (audio mode) [39] as objective metrics. Figure 3 shows the rate-distortion curves based on the bit-rate of the side information. We compare n-SBG and HE-AAC v1 at various bit-rates for the side information, both utilizing AAC-LC at either 13.0 kbps or 9.4 kbps as a core codec. For reference signals, full-band signals were low-pass filtered to match the bandwidth specifications of HE-AAC v1. When consuming similar bit-rates for the side information, the n-SBG consistently outperforms the HE-AAC v1 across all objective metrics. With respect to 2f-model MMS, n-SBG maintains comparable performance to the HE-AAC v1 while utilizing approximately half the bit-rate for the side information.

Figure 4 illustrates the results of an A/B preference test conducted with 14 participants, using randomly selected 9 audio samples (3 speech, 3 music, and 3 mixed) from the test set. The listening test follows the same experimental setup as the objec-

Table 1: Objective scores comparing Blind SBG and n-SBG

	NMR ↓	2f-model ↑	ViSQOL ↑
Core: AAC-LC @ 9.4 kbps	-1.74	28.34	2.52
+ Blind SBG	3.78	32.15	3.38
+ n-SBG (+2.57 kbps)	-2.80	46.04	4.38
Core: AAC-LC @ 13.0 kbps	-2.05	33.00	2.38
+ Blind SBG	1.51	37.47	3.34
+ n-SBG (+3.04 kbps)	-3.54	55.83	4.44

tive evaluation. Results show that n-SBG achieves higher preference scores compared to HE-AAC v1, even when the n-SBG allocates about half the bit-rates for the side information. Furthermore, the preference for n-SBG over SBR becomes more dominant as the bit-rate of the core codec decreases, indicating the effectiveness of n-SBG in low bit-rate conditions.

4.4. Ablation Study and Discussion

In Table 1, we compare the objective evaluation results of the core codec output, the n-SBG output, and the output from n-SBG without side information, referred to as Blind SBG. The Blind SBG significantly underperforms n-SBG and shows even worse NMR than the core codec output, indicating severe audible noise is introduced. These results highlight the necessity of side information for generating high-frequency components of complex audio signals.

Despite the overall performance of n-SBG surpassing SBR, its effectiveness varies depending on the characteristics of audio signals. n-SBG generates more realistic and vibrant transient components than the SBR but often struggles with generating tonal components with prominent harmonic structures. For future research, we will explore more effective conditioning strategies and high-frequency generation methods while also aiming to develop a system adaptive to various core codecs with different bit-rates.

5. Conclusion

In this work, we propose neural Spectral Band Generation (n-SBG), an alternative approach to rule-based Spectral Band Replication (SBR). The proposed method reconstructs high-frequency components by encoding and transmitting parametric information, which the SBG decoder efficiently utilizes for generation. Experimental results demonstrate that the n-SBG significantly outperforms the SBR at comparable bit-rates, with particularly notable efficiency gains observed in low bit-rate scenarios. However, n-SBG struggles with generating complex tonal components and requires training separate models for different bit-rates. Developing a unified system adaptive to various core codecs and bit-rates should be investigated in future works.

6. References

- [1] M. Bosi and R. E. Goldberg, *Introduction to digital audio coding and standards*. Springer Science & Business Media, 2002.
- [2] T. Painter and A. Spanias, “Perceptual coding of digital audio,” *Proceedings of the IEEE*, vol. 88, no. 4, pp. 451–515, 2000.
- [3] J. D. Johnston, “Estimation of perceptual entropy using noise masking criteria,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1988, pp. 2524–2527.
- [4] H. Fastl and E. Zwicker, *Psychoacoustics: facts and models*. Springer Science & Business Media, 2006, vol. 22.
- [5] *Information technology — Coding of moving pictures and associated audio for digital storage media at up to about 1,5 Mbit/s, Part 3: Audio*, ISO/IEC 11172-3, 1993.
- [6] M. Bosi, K. Brandenburg, S. Quackenbush, L. Fielder, K. Akagiri, H. Fuchs, and M. Dietz, “ISO/IEC MPEG-2 advanced audio coding,” *Journal of the Audio engineering society*, vol. 45, no. 10, pp. 789–814, 1997.
- [7] E. Larsen and R. M. Aarts, *Audio bandwidth extension: application of psychoacoustics, signal processing and loudspeaker design*. John Wiley & Sons, 2005.
- [8] *Information technology — Coding of Audio-Visual Objects — Part 3: Audio*, ISO/IEC 14496-3, 2005.
- [9] T. Ziegler, A. Ehret, P. Ekstrand, and M. Lutzky, “Enhancing mp3 with SBR: Features and capabilities of the new mp3PRO algorithm,” in *Audio Engineering Society Convention 112*. Audio Engineering Society, 2002.
- [10] J.-M. Valin, G. Maxwell, T. B. Terriberry, and K. Vos, “High-quality, low-delay music coding in the opus codec,” in *Audio Engineering Society Convention 135*. Audio Engineering Society, 2013.
- [11] M. Dietz, L. Liljeryd, K. Kjørting, and O. Kunz, “Spectral band replication, a novel approach in audio coding,” in *Audio Engineering Society Convention 112*. Audio Engineering Society, 2002.
- [12] S. Han and J. Lee, “Nu-wave 2: A general neural audio upsampling model for various sampling rates,” in *Proc. Interspeech*, 2022, pp. 4401–4405.
- [13] H. Liu, K. Chen, Q. Tian, W. Wang, and M. D. Plumbley, “AudioSR: Versatile audio super-resolution at scale,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 1076–1080.
- [14] Y. Li, M. Tagliasacchi, O. Rybakov, V. Ungureanu, and D. Roblek, “Real-time speech frequency bandwidth extension,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 691–695.
- [15] Y.-X. Lu, Y. Ai, H.-P. Du, and Z.-H. Ling, “Towards high-quality and efficient speech bandwidth extension with parallel amplitude and phase prediction,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 33, pp. 236–250, 2025.
- [16] P. Ekstrand *et al.*, “Bandwidth extension of audio signals by spectral band replication,” in *Proc. 1st IEEE Benelux Workshop on Model based Processing and Coding of Audio (MPCA-2002)*, vol. 6, 2002.
- [17] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, “Soundstream: An end-to-end neural audio codec,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 495–507, 2021.
- [18] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, “High fidelity neural audio compression,” *Transactions on Machine Learning Research*, 2023.
- [19] R. Kumar, P. Seetharaman, A. Luebs, I. Kumar, and K. Kumar, “High-fidelity audio compression with improved RVQGAN,” *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [20] Y. Ai, Y.-X. Lu, X.-H. Jiang, Z.-Y. Sheng, R.-C. Zheng, and Z.-H. Ling, “A low-bitrate neural audio codec framework with bandwidth reduction and recovery for high-sampling-rate waveforms,” in *Proc. Interspeech*, 2024, pp. 1765–1769.
- [21] A. Biswas and D. Jia, “Audio codec enhancement with generative adversarial networks,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 356–360.
- [22] J. Deng, B. Schuller, F. Eyben, D. Schuller, Z. Zhang, H. Franco, and E. Oh, “Exploiting time-frequency patterns with LSTM-RNNs for low-bitrate audio restoration,” *Neural Computing and Applications*, vol. 32, no. 4, pp. 1095–1107, 2020.
- [23] K. Gupta, S. Korse, B. Edler, and G. Fuchs, “A DNN based post-filter to enhance the quality of coded speech in MDCT domain,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 836–840.
- [24] S. Hwang, Y. Cheon, S. Han, I. Jang, and J. W. Shin, “Enhancement of coded speech using neural network-based side information,” *IEEE Access*, vol. 9, pp. 121 532–121 540, 2021.
- [25] J. Lin, K. Kalgaonkar, Q. He, and X. Lei, “Speech enhancement for low bit rate speech codec,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 7777–7781.
- [26] H. J. Nussbaumer and M. Vetterli, “Pseudo quadrature mirror filters,” in *Proceedings International Conference on Digital Signal Processing*, 1984.
- [27] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [28] M. Tagliasacchi, Y. Li, K. Misiunas, and D. Roblek, “Seantet: A multi-modal speech enhancement network,” in *Proc. Interspeech*, 2020, pp. 1126–1130.
- [29] S. Birnbaum, V. Kuleshov, Z. Enam, P. W. W. Koh, and S. Ermon, “Temporal FiLM: Capturing long-range sequence dependencies with feature-wise modulations,” *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [30] J. H. Lim and J. C. Ye, “Geometric gan,” *arXiv preprint arXiv:1705.02894*, 2017.
- [31] A. Van Den Oord, O. Vinyals *et al.*, “Neural discrete representation learning,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [32] J. Kong, J. Kim, and J. Bae, “HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 17 022–17 033, 2020.
- [33] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, “FSD50k: an open dataset of human-labeled sound events,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 829–852, 2021.
- [34] Z. Rafii, A. Liutkus, F.-R. Stöter, S. I. Mimilakis, and R. Bittner, “The MUSDB18 corpus for music separation,” 2017.
- [35] C. Veaux, J. Yamagishi, K. MacDonald *et al.*, “CSTR VCTK corpus: English multi-speaker corpus for CSTR voice cloning toolkit,” *University of Edinburgh. The Centre for Speech Technology Research (CSTR)*, vol. 6, p. 15, 2017.
- [36] M. Neuendorf, P. Gournay, M. Multrus, J. Lecomte, B. Bessette, R. Geiger, S. Bayer, G. Fuchs, J. Hilpert, N. Rettelbach *et al.*, “Unified speech and audio coding scheme for high quality at low bitrates,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2009, pp. 1–4.
- [37] K. Brandenburg and T. Sporer, “‘NMR’ and ‘Masking Flag’: Evaluation of quality using perceptual criteria,” in *Audio engineering society conference: 11th international conference: test & measurement*. Audio Engineering Society, 1992.
- [38] T. Kastner and J. Herre, “An efficient model for estimating subjective quality of separated audio source signals,” in *2019 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*. IEEE, 2019, pp. 95–99.
- [39] M. Chinen, F. S. Lim, J. Skoglund, N. Gureev, F. O’Gorman, and A. Hines, “ViSQOL v3: An open source production ready objective speech and audio metric,” in *2020 twelfth international conference on quality of multimedia experience (QoMEX)*. IEEE, 2020, pp. 1–6.